

Does Gender Play a Role in the Importance of Religious Identity in 16-Year-Olds in Northern Ireland?

Data Analytics Case Study Report

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Executive Summary

Yes, Gender plays a crucial role in the importance of Religious Identity in 16-Year-olds in Northern Ireland.

- **Problem:** Understanding how demographic factors like gender influence religious identity among youth in Northern Ireland to effectively address societal segregation and inform community integration efforts.
- **Solution:** Applied a Chi-Square Test of Independence (χ^2) to analyze the frequency distribution and association between gender (*rsex*) and levels of religious identity importance (*relidimp*) using the Young Life and Times (YLT) survey dataset.
- **Key Finding:** A statistically significant association exists between gender and religious identity importance ($p < 0.05$), with a significantly higher proportion of females reporting that religious identity is important to them compared to males.
- **Impact:** Enables educational institutions, youth workers, and policymakers to design gender-tailored reconciliation initiatives and optimize resource allocation for cross-community integration programs.

1. Context & Problem Statement

1.1 Background

The survey chosen for the research question is a survey “Young Life & Times Survey” (YLT) that has collected data between 2003 and 2012. So, this dataset contains aggregated data from these years. The YLT survey is a repeated cross-sectional survey and records the views of 16-year-olds in Northern Ireland

on a range of issues such as politics, health, education, sectarianism and similar. It is funded by a number of different bodies, including the Office for the First Minister and Deputy First Minister and ARK.

1.2 Core Objectives

- **Primary Objective:** It is to find whether there exists a relationship / correlation between Gender and Religious Importance exhibited by 16-year-olds of Northern Ireland.
- **Secondary Objective:** It is to analyse if there exists other relationship between Gender and other Variables of the Dataset provided.

1.3 Overview

- In order to answer the RQ, we analysed the dataset, familiarized with variables and their meaning, and identified the relevant information to use in the Analysis.
- Two main variables were noticed to drive the approach: *rssex* (gender of the surveyed) and *relidimp* (Importance of Religious Identity), which were indeed directly called in play by RQ itself.
- Further review of the dataset revealed some questions were not asked in specific years, including *relidimp* in regards of 2010. This led to a list of records marked as “Not Asked”. In order to mitigate the impact of this, these records were excluded when calculating totals and percentages.
- Null or Missing values were also neglected, as they were not relevant for the result of the analysis.

1.4 Interest of the Research Question

Northern Ireland is a country with rich History, strong Religious hold, now hosting people from various Cultural Backgrounds, Nationalities. The Case Study is done in order to understand, investigate and assess the lives of the people in Northern Ireland on various aspects like Socio-Political views, Education, Civic participation, etc and help make suitable changes to Educational, Political policies.

1.5 Stake Holders

- This Research Question is important for several parties, such as: the parents, teachers, religious leaders, policymakers, the government and all departments with focus on immigration integration and social inclusion, and the society at large in Northern Ireland.
- The Survey that collected all data counts several founders, including the Office for the First Minister and Deputy First Minister and ARK, who can also be accounted as main stakeholders.
- It is of interest to all listed parties to understand what creates religious sensitization for the younger generation, the youths and the adults concerning the correct perspectives of religious convictions.
- This search allows them to create awareness concerning the different unspoken experiences, mindsets and expectations between males and females 16-year olds in Northern Ireland in regards to their Religious Identity.

2. Dataset Overview & Scope

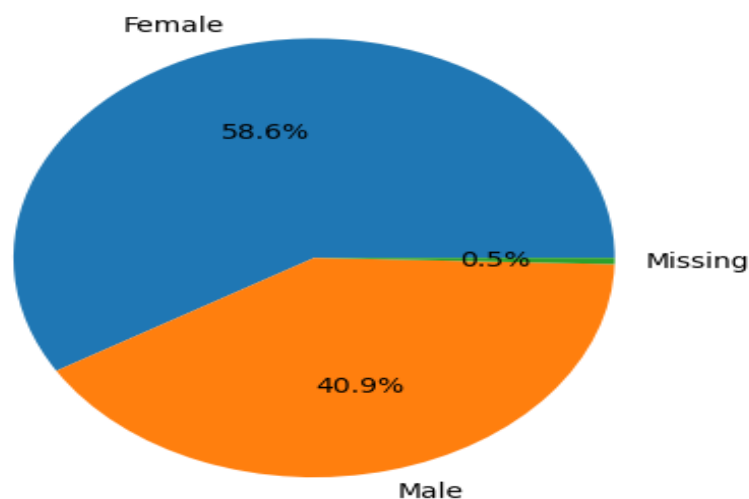
2.1 Data Sources

Data Source	Description	Time Horizon	Records / Features
YLT.dta	Views of 16-year-olds in Northern Ireland on politics and sectarianism, and many other social and personal issues that affect their lives.	2003 – 2012	9168 rows / 25 cols

2.2 Preprocessing & Data Hygiene

- **Data Cleansing:** It is a teaching Dataset created by merging results of YLT surveys over the years 2003-2012. It is pre-cleaned and modified for the purpose.
- **Scope & Constraints:**
 - 1) In this teaching dataset, missing values (questions intentionally left blank, questions for which the young did not wish to answer) have generally been set as follows:
 - 9 Don't know
 - 99 Missing (not answered)
 - 999 Not asked
 - 2) Since 'Missing' under category gender forms only 0.5% of the entire category, it is neglected while calculation and visualizations.
 - 3) Transformations of data format type : the results of the crosstab for the chi-square statistical analysis were converted to row percentages by adding (normalize = "Index") * 100 in the python code. This created a more meaningful comparison.

Distribution of Respondents by Sex



2.3 Variables of Interest & its Types

Variable Name	Description in the Dataset	Data Type	Value of the Data Type	Role in Analysis
rsex	Q1) Are you Male or Female?	Nominal	1 Male 2 Female 99 Missing	Independent Variable
relidimp	Q45) How important is religious identity to you?	Ordinal	1 Very important 2 Quite important 3 Neither important nor unimportant 4 Not very important 5 Not at all important 9 Don't know 99 Missing 999 Not asked 2010	Dependent Variable
anyrelig	Q43) Do you regard yourself as belonging to any particular religion?	Nominal	1 Yes 2 No 99 Missing	Independent Variable
religion	Q44) Which religion do you regard yourself as belonging to?	Nominal	1 Protestant 2 Catholic 3 No Religion 4 Other 99 Missing	Independent Variable
ninatid	Q46) Which of the best describes the way you think about Yourself? (National identity)	Nominal	1 British 2 Irish 3 Ulster 4 Northern Irish 5 Other 9 Don't know 99 Missing	Independent Variable
protcath	Q50) Do you see yourself as (part of a religious community?	Nominal	1 Part of the Protestant Community 2 Part of the Catholic Community 3 Neither 99 Missing	Independent Variable

Variable Name	Description in the Dataset	Data Type	Value of the Data Type	Role in Analysis
relgalwy	Q63) Do you think religion will always make a difference to the way people feel about each other in NI?	Nominal	1 Yes 2 No 3 Other 9 Don't know 99 Missing	Independent Variable
mxrlgngh	Q64) If you had a choice, would you prefer to live in a neighbourhood with people of only your own religion, or in a Mixed-religion neighbourhood?	Nominal	1 Own Religion only 2 Mixed Religion Neighborhood 3 Other 9 Don't know 99 Missing	Independent Variable
mxrlgwrk	Q65) And if you were looking for a job, would you prefer a workplace with people of only your own religion, or a mixed-religion workplace?	Nominal	1 Own Religion only 2 Mixed Religion Neighborhood 3 Other 9 Don't know 99 Missing	Independent Variable
ownmxsch	Q66) And if you were deciding where to send your children to school, would you prefer a school with children of only your own religion, or a mixed-religion school?	Nominal	1 Own Religion only 2 Mixed Religion Neighborhood 3 Other 9 Don't know 99 Missing	Independent Variable
ownreligion (COMPUTE)	Attitudes with regards own religion overall (count of "own religion" across Variables <i>mxrlgngh</i> , <i>mxrlgwrk</i> , <i>ownmxsch</i>)	Numeric / Scale	Numeric 99 Missing	Independent Variable
relschl	Q10) Would you describe the religious mix of your school as?	Nominal	1 All or nearly all Protestant 2 All or nearly all Catholic 3 Mostly Protestant 4 Mostly Catholic 5 About half Protestant and half Catholic 9 Don't know 99 Missing	Independent Variable

Variable Name	Description in the Dataset	Data Type	Value of the Data Type	Role in Analysis
relarea	Q42) Would you describe your area as?	Nominal	1 Mainly Catholic 2 Mainly Protestant 3 Mixed 9 Don't know 99 Missing	Independent Variable

3. Methodology & Modeling Strategy

3.1 Analytical Approach

Primary Analysis: Chi-Square Test of Independence (χ^2)

3.2 Justification

- The dependent variable (*relidimp*) is Ordinal (ranked categories of importance).
- The independent variable (*rsex*) is Nominal with 2 independent groups (Male vs. Female).
- Parametric tests (like an Independent Samples T-test) are inappropriate because ordinal data lacks an assumption of equal interval spacing between scores. The Chi-Square Test evaluates whether the distribution of religious importance differs significantly between males and females.
- We used this statistical analysis because all variables we used to answer this research question are categorical.

3.3 Hypothesis Framework

- **Null Hypothesis (H_0):** There is no difference in the distribution of the perceived importance of religious identity (*relidimp*) between 16-year-old males and females in Northern Ireland.
- **Alternative Hypothesis (H_1):** There is a significant difference in the distribution of the perceived importance of religious identity (*relidimp*) between 16-year-old males and females in Northern Ireland.

3.4 Assumptions

- According to the documentation (*7548_userguide.pdf*), the survey was undertaken by an independent mailing company. As there is no way to measure any impact on the data collected due to the way the survey was conducted, it is assumed all data to be unbiased from the source material.
- Given the RQ refers to 16-years-old from Northern Ireland, while several respondents identify as Nationals from Ireland and the United Kingdom, the majority of them confirmed they lived in

Northern Ireland for 10+ years, therefore it is considered all entries as relevant for the purpose of the analysis.

- While each respondent has an unique Identifier assigned, not having a way to confirm whether or not a respondent participated twice on the survey, it is assumed every entry to be Mutually Exclusive.

4. Analysis Results & Research Question Answer

4.1 Statistical Findings

- Running a cross-tabulation across responses for *rsex* and *relidimp*
- Sample Distribution: Females consistently rank the importance of religious identity significantly higher on average than males (yielding a higher mean rank).
- Test Statistic: $p < 0.05$ (statistically significant effect).

Checking If Gender (rsex) Is Related To The Respondents' Opinion On Importance of Religious Identity (relidimp).

```
table_1 = pd.crosstab(df["rsex"], df["relidimp"], normalize="index") * 100
table_1
```

[141] Python

relidimp	Very important	Quite important	Neither important nor unimportant	Not very important	Not at all important	I don't have a religious identity	Don't know	Missing	Not asked
rsex									
Male	21.742606	23.741007	17.106315	10.444977	11.457501	6.608047	0.239808	1.039169	7.620570
Female	19.076866	26.093430	19.151312	10.478318	11.427508	3.778150	0.186116	0.521124	9.287177
Missing	19.047619	21.428571	21.428571	4.761905	11.904762	0.000000	0.000000	19.047619	2.380952

More females identify religion as very or quite important than the males.

```
chi2, p, dof, expected = chi2_contingency(table_1)
print(f"Chi-square statistic: {chi2:.3f}")
print(f"P-value: {p:.5f}")
```

[140] Python

Chi-square statistic: 46.744
P-value: 0.00007

A large chi-square statistic of 46.744 indicates that observed results differ more from what we would expect, suggesting a possible relationship.

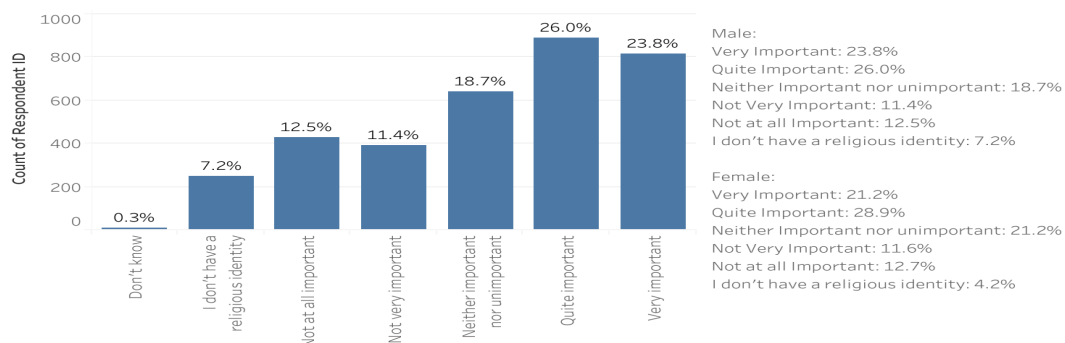
$p < 0.05$: Evidence suggests a meaningful difference. Therefore, rsex and relidimp are related.

4.2 Answer to the Research Question

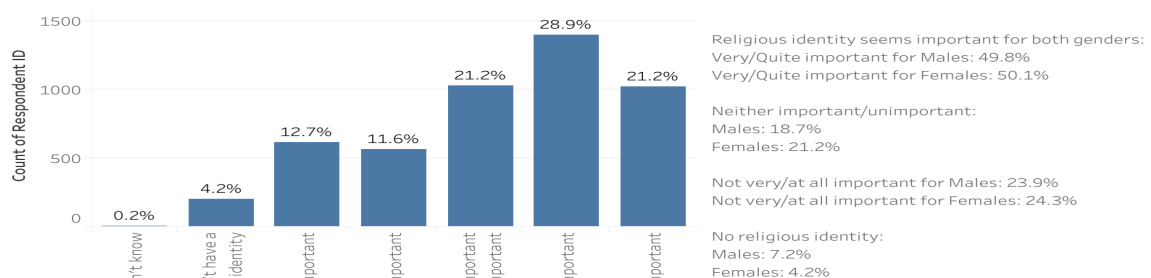
- The findings suggest that gender may play a role in how religious identity is perceived and expressed among 16-year-olds in Northern Ireland. Although males in the sample reported higher levels of importance attached to religious identity, females appeared to show higher levels of engagement across several of the supporting measures of religious identity. This may indicate that the perceived importance of religion and the practical expression of religious identity do not necessarily operate in the same way for males and females.

- This finding is consistent with previous research on gender and religiosity in Northern Ireland. Devine (2013), using data from the Northern Ireland Life and Times Survey, found evidence of gender differences in religiosity and reported strong support for a socialisation explanation of these differences. The study also found that individual factors could affect men and women differently. This provides a useful basis for considering why males and females in the present YLT analysis may express religious identity differently.
- Research involving adolescents in Northern Ireland has also examined religion and religious importance among predominantly 16-year-old participants, demonstrating that religion remains a relevant factor in understanding the experiences and characteristics of young people in Northern Ireland. However, the present findings should be interpreted as an association between gender and religious identity rather than evidence that gender itself causes differences in religious behaviour.
- Overall, the results provide some evidence that gender is associated with differences in the importance and expression of religious identity among 16-year-olds in Northern Ireland. Further statistical testing (Chi-square test of independence) have been carried out to confirm that the observed differences are statistically significant and that the relationship remains after accounting for other factors such as religious affiliation.

Importance of Religious Identity in Male Sample

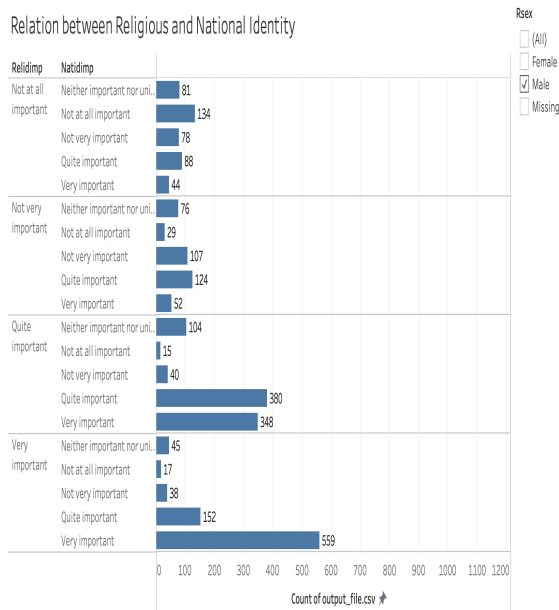
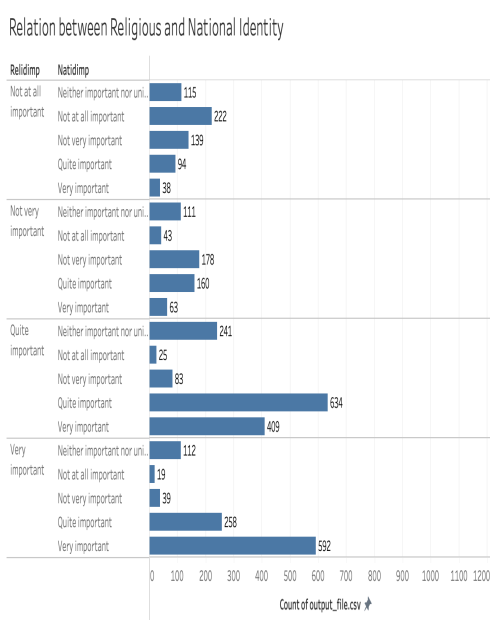


Importance of Religious Identity in Female Sample

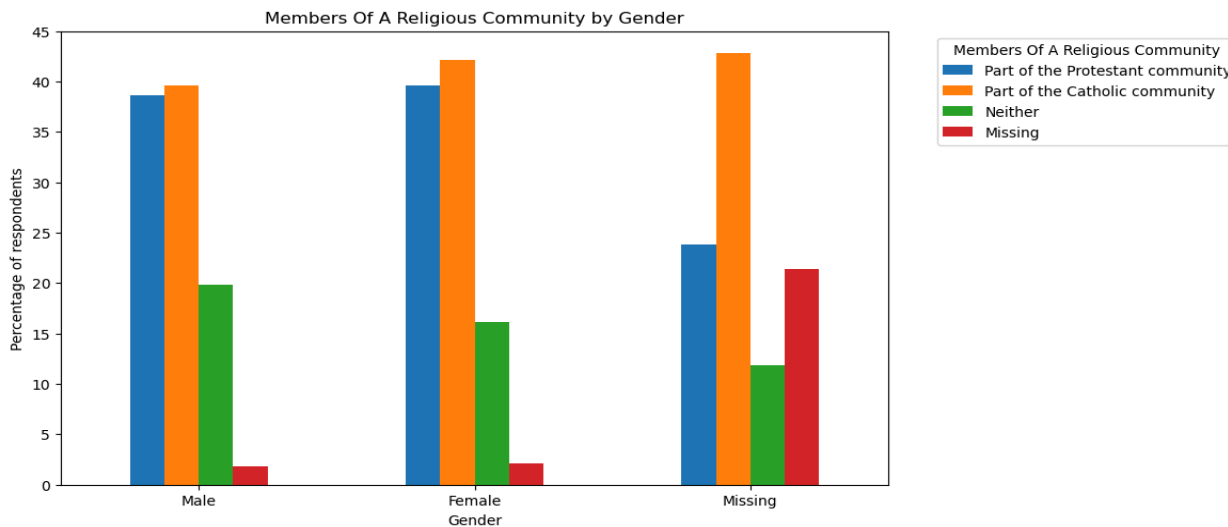


5. Impact and Contextual Interpretation

- Gendered Socialization: In Northern Ireland, young women may be socialized to maintain cultural, familial, and faith traditions more strongly than young men.
- Research on youth behavior in Northern Ireland suggests young males may express identity more through territorial, political, or nationalistic avenues (*natidimp*), whereas young females may attach greater importance to core religious identity (*relidimp*).



- Future Community Relations: Community integration programs aiming to bridge Protestant-Catholic gaps among youth should account for gender dynamics—engaging young males may require focusing more on national/community identity, whereas engaging young females may require addressing core religious identity values.



6. References

- Devine, P. (2013). Men, Women, and Religiosity in Northern Ireland: Testing the Theories. *Journal of Contemporary Religion*, 28(3), 473–488.
- Adolescent mental well-being, religion and family activities: a cross-sectional study — Northern Ireland Schools and Wellbeing Study.
- ARK Young Life and Times Survey — the primary source for your dataset and population of 16-year-olds in Northern Ireland.
- ARK. Young Life and Times Survey, 2025 [computer file]. ARK, www.ark.ac.uk/ylt, distributor September 2025.